Silvia Toonen

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Research positions

Oct 2025 - present	**Associate Professor** Anton Pannekoek Institute, University of Amsterdam, the Netherlands
Jan 2021 - Oct 2025	**Assistant Professor** Vidi-fellow laureate, Aspasia offer Anton Pannekoek Institute, University of Amsterdam, the Netherlands
Mar. 2019 - Dec 2020	**Lecturer** Institute of Gravitational Wave Astronomy, University of Birmingham, UK
Jan. 2017 - Feb. 2019	**Veni Fellowship** Dutch national fellowship for talented postdoctoral researchers Anton Pannekoek Institute, University of Amsterdam, the Netherlands
Jan. 2014 - July 2016	**Postdoctoral researcher** at Leiden Observatory, the Netherlands With Prof. Dr. Portegies Zwart at Leiden Observatory, the Netherlands
Sep. 2008 - May 2013	**PhD Astronomy** Thesis: The evolution of close binaries with white dwarf components With Prof. Dr. Nelemans at the Radboud Universiteit, the Netherlands

Extended work visits

Jan 2021 - Jan. 2024	**Honorary Lecturer** Institute of Gravitational Wave Astronomy, University of Birmingham, UK
Aug 2017 - Oct. 2017	**Visiting scolar** Collaboration with Dr. Shen and Prof. Dr. Kasen at the University of California at Berkeley, USA
Oct. – Dec. 2016	**Guest scientist** Collaboration with Dr. Perets at the Technion, Israel
Sep. - Nov 2013	**Postdoctoral researcher** With Prof. Dr. Nelemans at the Radboud Universiteit, the Netherlands
May - July 2013	**Work visit with Frye stipend** Collaboration with Prof. Dr. Gänsicke at the University of Warwick, UK
Aug. 2006 – Jan. 2007	**Research intern** With Prof. Dr. Beckman and Dr. Fathi at the Instituto de Astrofísica de Canarias, Tenerife, Spain.

Educational degree

2023 - 2024	**Teaching qualification** (BasisKwalificatie Onderwijs) *Teaching and learning centre, University of Amsterdam, The Netherlands*
2008-2013	**PhD, Astronomy** *Radboud University, Nijmegen, The Netherlands* Thesis: The evolution of close binaries with white dwarf components
2006-2008	**Master of Science, Astronomy** – Cum laude *Leiden University, Leiden, the Netherlands;* Resulting in two first-author papers on different topics

2002-2006 **Bachelor of Science in Astronomy**
Leiden University, Leiden, the Netherlands;

Fellowships & awards

- Nominated for Education Award 2025 at the University of Amsterdam
- (as PI) Grants from Dutch funding NOVA for the organisation of the workshop on "Advanced Simulations and Modeling in Astrophysics" in May 2025 (**€2500**)
- National Roadmap for Large-scale Research Infrastructures 2023, consortium member (**€12.000.000**)
- Incentive grant 2023 (**€190.000**)
- Marie-Curie 2023, host of Tjarda Boekholt
- Veni fellowship 2023, host of Eva La Place
- Host for AHEAD2020 visitor 2023 for Hila Glanz
- (as PI) Grant from Dutch funding agencies for the organisation of the BH workshop in June. 2023; NOVA (**€3920**)
- Host for HPC-Europe visitor 2022 for Andris Dorozsmai
- Vidi Fellowship 2021, Dutch national grant as PI, (**€800.000**)
- Veni fellowship 2021, host of Jan van Roestel
- Marie-Curie 2021 host of Tomer Shenar
- Nominated for Outstanding Teaching Award 2020 of the University of Birmingham
- Host for HPC-Europe visitor 2020-for Alexey Bobrick
- Newton-Wong 2020 host of Yan Gao
- Rubicon-Fellow 2019 host of Valeriya Korol
- Veni Fellowship 2016, Dutch national grant as PI, (**€250.000**)
- (as PI) Various grants from Dutch funding agencies for the organisation of the SNIa workshop in Feb. 2018; KNAW (**€5000**)**,** NOVA (**€4000**), NWO incidentele steun (**€2250**)
- (as PI) Various grants from Dutch funding agencies for the organisation of the triple workshop in Sep. 2018; KNAW (**€5000**)**,** NWO incidentele steun (**€2250**)
- Various grants from Dutch private funding agency LKBF for travel in 2008-present (**~€7000**)
- Travel grant from Texas A&M university in 2016 (**$1000**)
- Award for best talk of an early career scientist, conference in Lake Louise, Canada (2014)
- Frye Stipend 2011 for talented female PhD candidates (**€3500**)
- Cum laude degree, Master of Science, Astronomy, at Leiden University
- Scholarship from Leiden University Fund for a 6-month research internship at Instituto de Astrofísica de Canarias, Tenerife, Spain (2006)

Selected invited reviews, colloquia, seminars and talks

- >30 invited colloquia & seminars at major research institutions around the world including at Oxford, Cambridge and Harvard ITC.
- >30 invited talks & reviews at a variety of international conferences and workshops

Invited colloquia and seminars: Astron, The Netherlands, 2026; Cambridge IoA, UK, 2026; Paris, France, 2025; Indiana, USA, 2025; Dublin, Ireland, 2024; Sheffield, UK, 2023; AEI Hannover, 2022, Hannover, Germany; Boston University, 2022, Boston, USA; Heidelberg Joint Astronomical Colloquium, 2022, Heidelberg, Germany; MPG/AEI, 2021, Potsdam, Germany; UFRGS, 2021, Porto Alegre, Brazil; Leiden Observatory, 2021, Leiden, The Netherlands; JIVE, 2021, Dwingeloo, The Netherlands; Mullard Space Science Laboratory, 2021; Groningen, The Netherlands, 2021; Liverpool John Moores University, 2021; Exeter, UK, 2021; Warwick, UK, 2021; Carnegie-Mellon & Pittsburgh, USA, 2021; Caltech, USA, 2021; Geneva & Sauverny, Switzerland, 2020; Carnegie Observatories, USA, 2020; NRC-HIA, Victoria, Canada, 2020; JBCA & Manchester, UK, 2020; Royal Observatory, Edinburgh, UK,

	2020; Oxford, UK, 2020; Cambridge DAMTP, UK, 2019; Surrey, UK, 2019; London UCL, UK, 2019; Harvard ITC, USA, 2019; Nijmegen, The Netherlands, 2018; Southampton, UK, 2018; Leuven, Belgium, 2017; Lund, Sweden, 2017; Berkeley, USA 2017 (TAC Seminar); Haifa, Israel (2016).
Plenary speaker	LISA-NL, Amsterdam, 2024; LISA symposium 2022, Glasgow; NAC, Leiden, 2021; TRENDY III, Chicago, 2021;
Discussion panellist	O4 GW Lorentz workshop, Leiden, 2024; GWPAW 2024, Birmingham, 2024; NAC Leiden, 2021; GWPAW 2021, virtual, 2021;
Invited talks & Reviews:	3rd International Online Conference on Universe (IOCU) 2026, online, 2026; Transients, Padua, Italy, 2025; LGWA workshop, Rome Italy, 2024; Triples, Florence, Italy, 2024; N-body dynamics, Garching, Germany, 2024; LISA2050, online, 2024; Massive multiples, Leuven, Belgium, 2023; Alpha Centauri, Nice, France, 2023; Cambridge, 2022; Quy Nhon, Vietnam, 2022; LISA star symposium, virtual, 2022; GWPAW 2021, virtual, 2021; CEPO2021, virtual,2021; EAS SS7, Leiden, NL, 2021; EAS SS29, Leiden, NL, 2021; DDA, USA, 2021; Leuven/Bonn meeting, 2020; ESO, Garching, Germany, 2020; KITP, USA, 2019; Warsaw, Poland, 2018; Rome, Italy, 2018; Berlin, Germany, 2018; Birmingham, UK, 2017; Stockholm, Sweden (2x), 2017; Rome, Italy, 2017; Palermo, Italy, 2015; EWASS, Tenerife, Spain, 2015; Lake Louise, Canada, 2014 (award);
Talk:	Cambridge, UK, 2019; Garching, Germany 2017; Stockholm, Sweden, 2017; Forth Worth, USA, 2017; Dallas, USA, 2016; Haifa, Israel, 2015; Ulaan Baatar, Mongolia, 2014; Garching, Germany, 2012; Krakow, Poland, 2012; NAC, The Netherlands, 2012; Noordwijk, The Netherlands, 2012; NOVA meetings, The Netherlands, 2012 & 2013.
Poster:	Padua, Italy, 2011; Tübingen, Germany, 2010; Mykonos, Greece, 2010; Tucson, USA, 2010; NAC, The Netherlands, 2009 and 2010.
Workshops:	Leiden, The Netherlands, 2x in 2019; Sheffield, UK, 2018; Leiden, The Netherlands, 2018; Stockholm, Sweden, 2017; Leiden, The Netherlands, 2013; Leiden, The Netherlands, 2010; Leiden, The Netherlands, 2009; Cape Town, South-Africa, 2008; NOVA fall school, The Netherlands, 2008.

Host of long-term visitors

May-Jun 2025	Ana Escorza, Instituto de Astrofysics de Canarias, Spain
May-Jun 2025	Michael Abdul-Masih, Instituto de Astrofysics de Canarias, Spain
Jan 2025	Alexey Bobrick, Technion, Israel / Monash University, Australië
2024-ongoing	Zephyr Penoyre, Leiden Observatory, recurrent visitor
2023-ongoing	Devina Misra, Norwegian University of Science and Technology, recurrent visitor
May-June 2023	Luca Sciarini, University of Geneva, supported by LKBF & NOVA
Sep-Oct 2022	Hila Glanz, Technion, Israel, supported European Union's Horizon 2020 Programme under the AHEAD2020 project
April 2022	Matthias Fabry, Katholieke Universiteit Leuven
Jan-Mar 2022	Andris Dorozsmai, University of Birmingham, supported by HPC-Europe
2021-2024	Steven Rieder, University of Geneva, recurrent visitor
Nov. 2020	Alexey Bobrick, Lund Observatory, supported by HPC-Europe

Organisational skills

Scientific organisation:

2026 - present	**LISA Consortium Council member**
2026 - present	**Chair** of LISA Astrophysics Working Group (~500 international members)
2025 - present	**Chair** of LISA Astrophysics Working Group Project on planetary and stellar triples

2025 - present **Chair** of LISA Astrophysics Working Group project on comparing LISA UCB catalogues with observations
2025 - present **ET Collaboration Board member**
2025 - present **ET Research unit leader** University of Amsterdam
2025 - present **Founder and elected chair** of LISA member group LISA-Amsterdam
2023 - present **Chair** of GW-Space 2050 work package on Galactic compact binaries
Exploration of future GW mission in response to ESA's Voyage 2050 call
2023-2024 **Chapter coordinator** for ET blue book
2022-present **Core member** of the 4MOST Science team, white dwarf binary survey
2021-present **Core member** of the Einstein Telescope Observation Science Board, division 3&7
2020-2023 **Chapter coordinator** for LISA white book
2019-present **Chair** of LISA working package on 'Demographics, formation, evolution and electromagnetic counterparts of stellar-mass compact objects'.
2017-present **Core member** of the WEAVE Science team, white dwarf survey
2010-2014 **Principal investigator and team-lead** of PopCORN project
Extensive collaboration for comparison of various binary population synthesis codes. Paper published in A&A.
2003-2004 **Organiser** of scientific excursion of 9 days to Munich and Vienna for 40 students

Conference organisation:

2025 SOC: EAS S5 symposium on 'Binary Stellar Interactions and Their Outcomes', Cork, Ireland
2025 SOC: 'Advanced Simulations and Modeling in Astrophysics', Zwolle, The Netherlands
2025 Co-chair: 'Circumbinary systems formation and evolution', Florence, Italy
2024 SOC: 'Modest 24', Warsaw, Poland
2024 Co-chair: '360 degree approach to Common Envelope Evolution', Barcelona, Spain
2023 SOC: EAS S11 symposium on 'Stellar interactions: contact binary stars and common-envelope evolution', Krakow, Poland
2023 Co-chair: Workshop 'BH detections in the local group', Leiden, NL
2022 Co-chair: 3-month KITP workshop ' White dwarfs as probes', CA, USA
2022 SOC: GWPAW Conference 'GW physics and astronomy workshop', Melbourne, Australia
2022 SOC: EWASS support session for early career astronomers, Valencia, Spain;
2021 SOC: EWASS S8 special session on 'Stars on the pathway to become GW sources', Leiden, NL
2021 SOC: EWASS support session for early career astronomers, Leiden, NL
2021 Co-chair, KITP virtual mini workshop ' White dwarfs as probes', CA, USA
2021 SOC: Conference 'Triple evolution & dynamics III', ~200 participants, Chicago, USA
2020 SOC: 'Triple evolution & dynamics III', China, cancelled due to Coronovirus
2019 SOC: Conference 'Compact white dwarf binaries', ~100 participants, Armenia
2019 SOC: Conference 'Binaries in the universe', ~100 participants, Czech Republic
2019 SOC: Conference 'Stars on the run II', ~60 participants, Germany
2019 Co-chair: Conference 'Beginnings & ends of double white dwarfs', ~60 participants, DK
2019 Co-chair: Workshop 'Beginnings & ends of double white dwarfs', ~12 participants, DK
2019 Co-chair: EWASS special session 'Stellar multiplicity in the Gaia era', France
2019 Co-chair: Workshop 'Electron-Capture-Initiated Stellar Collapse', ~25 participants, NL
2018 **Chair:** Lorentz Centre Workshop 'Triple evolution & dynamics II', 55 participants, NL
2018 Co-chair: Workshop 'Observational signatures of Type Ia supernovae Progenitors III', ~55 participants, NL
2016 SOC: Conference 'Stars on the run', ~60 participants, Germany

*SOC = member of Scientific Organising Committee

Institutional responsibilities:

2024-present Founding member of committee for the promotion of Data science & Computational research & support

2023-present	Member of committee to develop the new bachelors program on 'Science, Technology & Innovation', representative of astronomy
2022-2023	Member of recruitment panel tenure track staff positions at Univ. of Amsterdam
2022-present	Organiser of NOVA's yearly career weekend for 3th year PhD students
2021-present	Co-chair of WiF (Women in the Faculty of Science) at Univ. of Amsterdam
2020-2021	Member of Equity, Inclusion and Diversity Committee at Univ. of Amsterdam
2019-2021	Organiser of astrophysics and space research seminar series
2019-2021	Founding organiser of PhD meet & greet with seminar speakers
2018-2021	Member of Equity, Inclusion and Diversity Committee at Univ. of Birmingham
2018-2019	Member of search committee for postdoctoral fellows at Univ. of Birmingham
2018-2021	Member of recruitment panel for PD and PhDs candidates at Univ. of Birmingham
2017-2019	Founding organiser of bi-weekly lunch meeting for postdocs Aim: foster the postdoc community and exchange ideas and experiences, regarding teaching & mentoring, job applications etc.
2014-2015	LEAPS 2015 (Leiden/ESA astrophysics Program for undergraduate Summer Students) REU-level internship for 22 students at Leiden Observatory, ESA and science-based companies (Airbus, TNO). Organisation included promotion of program, recruitment of candidate students and research supervisors, application procedure, housing, stipends, colloquium series etc.
2014-2015	Organisation and chair of weekly group meeting (2014-2015) on high-energy astrophysics

Teaching and mentoring experience

Mentoring and (co-) supervision of graduate and undergraduate students:
Resulting in theses and publications

- Primary supervisor of 4 PhD students (A. Parkosidis, C. Bruenech, F. Kummer, University of Amsterdam, A. Dorozsmai, Birmingham)
- Co-promotor of PhD student (F. Zainol, Birmingham)
- Host of 6 PD fellows (T. Boekholt, E. La Place, J. Van Roestel, T. Shenar, Amsterdam; V. Korol & Y. Gao, Birmingham)
- Primary supervisor of 10 masters students (P. Garcia Hoogland, J. Romero, G. Hanegraaf, T. Keizer, A. Katechakis, M. Stroet, A. Parkosidis, G. Simion, T. Konijn, K. Temmink (paper published in A&A), Amsterdam),
- Primary supervisor of 4 MPhys students (Birmingham)
- Primary supervisor of 4 bachelors studenten (G. Kokosky Deforchaux, C. Keijzer, L. Krebbers, R. Limmen, Amsterdam)
- Primary supervisor of 3 Astrovaria studenten (H. Falkena, S. Gandini, D. Agrawal, Amsterdam)
- Primary supervisor of 4 summer students (A. Das, M.A. Husain, S. González Zamora, M. Vilaxa Campos, Amsterdam; D. Bandopadhyay, Birmingham; B. Curd, Leiden)
- Co-supervisor of 6 masters students (C. Keijzer, K. Theodoridou, Q. Dong, F. Temmink, S. Luijten, Amsterdam; M. Bours, Nijmegen, paper published in A&A)
- Co-supervisor of 4 bachelors students (T. Vaessen, E. Van der Grijp, Amsterdam; H. Klein Woud, Leiden; R. Castelijn, Nijmegen)
- Mentor of 6 masters students

Former students that continued in academic positions:

Vilaxa Campos (PhD in Nijmegen, the Netherlands), Stroet (PhD in Armagh, Northern Ireland), Konijn (PhD in Delft, the Netherlands), Dorozsmai (PD in NAOJ, Japan), Parkosidis (PhD in Amsterdam, the Netherlands), K. Temmink (PhD in Nijmegen, the Netherlands), Curd (PhD at Harvard CfA, PD UTSA, USA), Bours (PhD in Warwick, PD in Valparaiso), Castelijn (PhD at Nikhef, the Netherlands).

Former postdoctoral group members that continued in academic positions:
la Place (prof. at KU Leuven), van Roestel (PD at ISTA, Austria), Shenar (prof. at Tel Aviv University), Korol (PD in MPA & Cambridge, prof. at SRON, The Netherlands).

Teaching & course developments:

Sept. - Oct. 2024	Guest lecturer for masters course 'Open problems in modern astronomy'
Aug. 2024	Lecturer at Niels Bohr International Academy, summer school on astrophysical dynamics of gravitating systems
2023-present	Development of new bachelors program 'Science, Technology and Innovation"
2021-present	Lecturer for masters course 'Stellar evolution and structure'
Jan.-Apr. 2020	Lecturer for bachelors course 'Introduction to Astronomy'
Oct.-Nov. 2019	Lecturer & supervisor at GrEAT program (Gravitational-wave Excellence through Alliance Training) on 'white dwarfs in the gravitational wave era'
June 2018	Guest lecturer for Dutch Astronomy Olympiad on 'binary stars & GWs'
Nov. 2015	Guest lecturer for university course 'Compact Object and Accretion'
2008-2010	Academic teaching assistant Courses on stellar evolution, asteroseismology, compact binaries, optics & telescopes. The responsibilities include teaching exercise classes and computer lab, development of new materials, grading
2003-2008	Head teacher and assistant teacher mathematics at preparation training for final exams at high school (2003-2008). The responsibilities include: • Assist with organising and preparing 3-day courses and its accompanying materials • Class wide explanation of the subjects and exercise classes. • Responsibility over 20 to 30 students and 4 to 6 assistant-teachers

Miscellaneous

Commissions of trust:

PhD committee member at the University of Amsterdam

- Ekaterina Leonova, Stefanie Fijma, Dion Linssen, Devarshi Choudhury, Uddipta Bhardwaj, Saugata Barat, Mitchell Stoop, Lieke van Son, Per Gunnar Valegard, Sarah Brands, Geert Raaijmakers, Kenzie Nimmo, Mathieu Renzo, Manos Zapartas.
- PhD committee member at other institutes
  Nicolaus Copernicus Astronomical Center, Torun (Ayush Moharana); University of California, Santa Cruz (Rosa Wallace Everson), University of Conception (Jose Manuel Pinto Pereira); Katholieke Universiteit Leuven (Gareth Banyard, Soetkin Janssens); Rochester Institute of Technology (Emily Wilson); University of Warwick (Elena Cukanovaite); University of Birmingham (Coen van Neijssel); Leiden University (Alex Rimoldi).
- IAU member
- Funding referee for ESO, STFC, Fondecyt, Binational Science Foundation (US-Israel)
- Scientific referee of the journals Nature, A&A, MNRAS, ApJ, MNRAS and the HST observatory
- Member of the Education Advisory Committee of Astronomy (2006-2008) and Physics & Astronomy combined (2005-2008) at Leiden Observatory

Computational expertise:

TRES (PI)	Innovative code for modelling triple star evolution
SeBa (developer)	Fast, parameterized code for modelling binary evolution
MESA (user)	Detailed code for modelling single stellar evolution
AMUSE (user)	Software framework for astrophysical simulations to apply and couple codes from different domains (e.g. stellar evolution, stellar dynamics, hydrodynamics)
Programming languages	C++ (fluent), Python (fluent), IDL (basic), Fortran-77 (basic), LaTex (fluent), SVN (fluent), Github (fluent)

Service:

- Founder and moderator of the first Facebook group dedicated to triple evolution and dynamics with the aim to connect researchers working on multiple systems on diverse scales and methods. Particularly important in the time of restricted travel (2020).
- Synthetic models of Milky Way double white dwarfs for predictions and testing of data analysis pipelines of LISA, Gaia etc. Moreover, my models are integrated in the official LISA data processing flow, and as such are used e.g. in the ESA application for the L3 missions of the Cosmic Vision Programme in 2017 (arXiv:1702.00786)

Public outreach:
- News item on main commercial TV network, Editie NL on RTL4, (March 2024)
- Several press interviews (Kijk, astronomy.com, Trouw)
- Radio Swammerdam; science radio show (February 2024)
- Development of outreach package sterrestrijd (2023-)
- Contributed to development of outreach package 'Astronomy as a Field: A Guide for Aspiring Astrophysicists' (2023)
- Contributed to development of outreach package 'Stellar evolution Scavenger Hunt' in the Program Cookbook for *Girls STEAM Ahead with NASA* and *NASA's Universe of Learning* (2023)
- Personal interview (Zenit, 2023)
- Author of popular article for scientific magazine 'Zenit'
- General outreach through popular lectures at public observing nights, university open days, intro to physics to young girls, associations for weather and astronomy etc.
- Lecture at career day for masters students (November 2017)
- OC of renovation of exhibits at & fundraising for the Old Observatory Museum in Leiden (2014-2015)

Observing experience:
- Photometric Survey of the Milky Way at the INT (July 2010, November 2009, July 2009)
- AMCVn spectroscopy at the WHT (November 2009)

Language skills: Dutch (native), English (fluent), German (reasonable), Spanish (reasonable)